PROS: A Proactive Multimodal Refinement Agent for Scientific Posters

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Abstract

Human-interactive generation and editing systems can produce strong first drafts, yet users often still struggle earlier in the loop: identifying what should be changed and how to express it as an actionable edit. We study this problem in the context of scientific posters, a structured visual artifact that couples layout, multimodal evidence, and source-grounded communication. We present ***PROS****, a proactive multimodal refinement framework that goes beyond reactive instruction following by helping users surface and repair under-specified improvement needs. Given a poster draft and its source paper, PROS supports human–AI collaborative, iterative refinement by diagnosing optimization-worthy deficiencies, assigning them to executable refinement stages, and applying grounded edits over editable poster states. The current workshop version centers on a compact six-class executable taxonomy spanning structural validity, space/layout reallocation, and content-level repair. On a preliminary editable benchmark, PROS achieves a macro diagnosis F1 of* ***82.0%****, an average repair resolution of* ***73.0%****, and a pilot holistic before/after gain of* ***29.1%****. These preliminary results position proactive poster refinement as a controllable multimodal visual editing workflow for human-interactive refinement of scientific artifacts, rather than instruction execution alone.*

1. Introduction

Recent paper-to-poster systems have made major progress on generating visually plausible first drafts from scientific papers [2, 8, 10, 11, 18, 19, 21]. Some of the newest systems even produce editable outputs such as PPTX, making downstream refinement operationally feasible. However, a generated draft is rarely a finished poster. Users still revise drafts to improve reading flow, rebalance sections, foreground stronger evidence, and repair layout-side effects.

This creates an emerging opportunity: to move from one-shot poster generation toward *poster improvement* as a first-class goal. But improvement itself is not only an edit-execution problem. In many cases, users do not know what is wrong in a precise enough way to specify the right edit. A blank region may look like a simple layout issue but actually stem from weak section allocation or missing content. A section may feel unconvincing because the poster under-communicates the source paper's problem, mechanism, or evidence. Users therefore face two coupled difficulties: identifying *what* should be edited and determining *how* to edit it.

This challenge reflects a broader limitation of reactive AI assistance. Recent work on proactivity argues that users often operate under incomplete understanding and cannot fully articulate what is missing or worth surfacing in advance [5]. Scientific poster refinement is a concrete instance of this setting: a user may feel that a poster is unclear, imbalanced, or under-explanatory, yet still be unable to specify what intervention would actually improve it. Helping effectively therefore requires more than waiting for explicit instructions; it requires helping the user discover and formulate the right edits.

Existing poster-editing and neighboring document-editing work has primarily framed refinement as instruction following or post-processing: the user specifies an edit, and the system grounds or executes it over a document, slide deck, or layout [4, 12, 13, 17]. Recent work on structured design editing also highlights that poster- and webpage-like artifacts require global layout reasoning beyond local object-level edits [16]. Among current systems, APEX is the closest poster-editing setting we are aware of, but it remains fundamentally interactive: the system provides an editing platform that follows user-specified instructions [13]. We study a different setting. Given a poster draft and its source paper, can a multimodal system determine *what* is worth improving before the user fully specifies the edits, infer *why* those issues matter, and carry out grounded changes while preserving structural validity?

This setting also expands the scope of productivity research. Many prominent demonstrations of generative-AI productivity gains focus on writing, customer support, or coding [1, 7, 9], while proactive-agent research has mainly centered on dialogue or task assistance [3, 6, 20]. We instead study *artifact-centric productivity* over structured two-

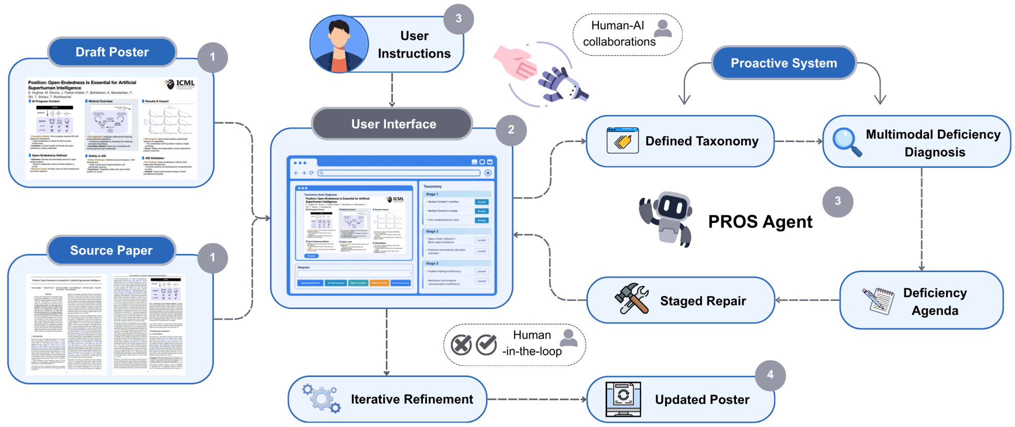


Figure 1. Overview of PROS as a proactive multimodal refinement agent over editable poster states. The system combines grounded editing over PPTX objects, multimodal deficiency diagnosis from poster–paper inputs, and fixed stage-wise refinement with re-rendering and re-diagnosis.

dimensional scientific communication artifacts. In this setting, the goal is not merely to produce another draft or respond to a fully specified request, but to help users discover under-specified improvement needs and convert them into grounded repairs over editable poster states.

We formulate this problem as **proactive scientific poster optimization**. PROS is designed as a *proactive collaborative multimodal refinement agent* for scientific posters. Rather than replacing the user, it surfaces actionable deficiencies, relates them back to the source paper, proposes stage-wise grounded refinements, and supports user-directed execution over editable poster states. PROS supports both sides of human–AI collaboration: when users know what they want changed, it can support interactive editing; when they do not know what is wrong or how to fix it, it can proactively identify and resolve optimization-worthy deficiencies. Conceptually, PROS treats proactive assistance not as "editing before being asked," but as helping users identify what they actually need, why it matters, and how it should best be repaired. For human-interactive generation, this frames poster refinement as a controllable multimodal visual editing workflow: the user and system iteratively refine a structured artifact, while the system helps discover and operationalize edits rather than merely execute explicit commands.

2. PROS: Staged Multimodal Poster Refinement

PROS combines three components: (i) a grounded editing substrate over editable poster objects, (ii) a compact executable taxonomy with multimodal deficiency diagnosis, and (iii) a stage-wise refinement policy that converts diagnosed deficiencies into grounded edits over editable poster states.

Grounded editing substrate. The system operates over editable poster drafts, with synchronized mappings between semantic blocks, underlying PPTX objects, and rendered visual regions. This representation supports deterministic grounded edits while still allowing multimodal diagnosis from the rendered poster. The same substrate supports both direct user editing and system-driven proactive refinement, enabling iterative updates over explicit poster states rather than full regeneration after each change. This is important because proactive assistance in our setting must go beyond critique: once a missing or weakly specified need is identified, the system must operationalize it into concrete changes over the editable artifact.

Executable core taxonomy. Rather than maintaining a large flat inventory of poster flaws, PROS uses a compact taxonomy designed for executable refinement. The current workshop version focuses on six core classes organized into three stages, summarized in Table 1. We select these classes as an executable core because they span three necessary layers of scientific poster refinement: structural validity, layout-level readability, and source-grounded scientific communication. This choice aligns with poster-design guidance emphasizing clear organization, readable layout, succinct content, and effective communication of study objectives, methods, findings, and implications [14, 15]. Each class is recurrent, observable from poster–paper inputs, repairable over editable poster states, and evaluable through before/after resolution. Our broader taxonomy development spans 18 deficiency classes derived from mixed-source poster annotation and analysis; the remaining classes serve as a broader diagnosis inventory for future repair policies. The full taxonomy is provided in Appendix A.3, with executable diagnosis and repair details summarized in Appendix A.6.

Table 1. Executable core taxonomy used in the workshop version.

Stage	Taxonomy classes
Stage 1	Overflow / out-of-bounds; overlap / collision
Stage 2	Excessive whitespace / blank-region imbalance; section allocation / layout imbalance
Stage 3	Content insufficiency; content redundancy / noise / overload

Multimodal deficiency diagnosis. Given a poster and its source paper, PROS identifies structured issue candidates using deterministic layout checks, section-level structural cues, paper-aware semantic signals, and multimodal aggregation over text and layout. This source-grounded design is central to the problem setting: many relevant deficiencies are not reducible to surface-level visual cleanup. A poster may look plausible while still omitting central problem framing, obscuring the method's mechanism, or failing to foreground the strongest evidence from the source paper. Conversely, a visible issue such as whitespace may reflect deeper content-allocation or communication problems. PROS therefore diagnoses not only what looks wrong on the artifact, but also what is missing, weak, or misallocated relative to the source. This allows the system to function as a collaborator in problem finding, not only a follower of pre-specified instructions.

Stage-wise refinement. PROS does not execute one flat edit list. It first resolves hard structural failures, then reallocates space and layout, and only then repairs remaining communication gaps. This ordering matters because the system is not merely applying local fixes; it is converting diagnosed needs into executable repair plans. Validity failures destabilize later edits, while blank regions and weak section allocation should be handled before deeper paper-grounded content refinement. This stage structure reflects a practical dependency order for turning under-specified user needs into grounded interventions. It also supports iterative stage-wise refinement: after each stage, the poster is re-rendered and re-diagnosed, allowing later stages to operate on the updated artifact state rather than on the original draft.

Formally, PROS iterates over editable poster states

$$P^{(\ell+1)} = \mathrm{Apply}\Big(P^{(\ell)}, A^{(\ell)}\Big), \tag{1}$$

where $P^{(0)}$ is the initial poster and $A^{(\ell)}$ is the active refinement agenda for stage ℓ. Each issue z_i is assigned to one of three refinement stages, $\lambda(z_i) \in \{1, 2, 3\}$, and PROS executes Stage 1, then Stage 2, then Stage 3, re-rendering and re-diagnosing after each stage.

3. Preliminary Benchmark and Results

We present a workshop benchmark and evaluation setting designed to make the diagnosis-to-refinement loop concrete enough for quantitative study while keeping the executable scope aligned with the current system implementation.

Dataset construction and scope. Our broader data collection is organized around *poster–paper pairs*, where each poster is paired with its original source paper. The current collection draws source papers from ICML, ICLR, CVPR, and NeurIPS across 2024 and 2025, with coverage balanced across the two years. It includes three data categories: 36 PosterGen-generated drafts, 36 Paper2Poster-generated drafts, and 36 human-made posters collected from conference materials. For the generated subsets, we run the corresponding paper-to-poster generation pipelines on selected papers and use the resulting poster drafts in our benchmark, rather than treating prior outputs as static artifacts. Additional dataset construction details are provided in Appendix A.1.

Taxonomy construction. These poster–paper pairs serve two roles. First, we use generated drafts and human-made posters together to support deficiency analysis and taxonomy construction. We perform both human annotation and VLM-assisted annotation over these materials, then consolidate recurring issues into a broader 18-class deficiency taxonomy. Second, for the present workshop submission, we restrict quantitative end-to-end evaluation to the 6-class executable subset shown in Table 1. This restriction reflects current implementation scope rather than the absence of a broader diagnosis inventory. Further details are provided in Appendix A.2.

Current experimental subset. The current quantitative experiments are conducted on an editable subset of 16 model-generated poster–paper pairs: 8 PosterGen drafts and 8 Paper2Poster drafts. This subset supports grounded diagnosis-and-repair evaluation over editable machine-generated posters, while human-made posters currently support taxonomy development and broader deficiency analysis. Details of the current experimental subset are provided in Appendix A.4.

Diagnosis benchmark. The diagnosis track evaluates whether PROS correctly identifies the six executable taxonomy classes. We manually annotate the presence or absence of each class for every poster in the evaluation subset and treat these annotations as the gold standard. System predictions are binarized at the class level, where a detected class is counted as 1 and an undetected class as 0. We report per-class precision, recall, and F1, together with macro-F1 over the six classes. Full evaluation protocol details are provided in Appendix A.5.

Table 2. Preliminary diagnosis performance on the current 16-case mixed subset (8 PosterGen + 8 Paper2Poster).

Stage	Taxonomy	Precision	Recall	F1
Stage 1	Overflow / out-of-bounds	0.88	0.88	0.88
	Overlap / collision	0.52	0.72	0.61
Stage 2	Whitespace / blank-region imbalance	1.00	1.00	1.00
	Section allocation / layout imbalance	1.00	1.00	1.00
Stage 3	Content insufficiency	0.73	0.69	0.71
	Content redundancy / noise / overload	0.76	0.70	0.73
Macro average		–	–	0.82

Table 3. Preliminary repair performance on the current 16-case mixed subset (8 PosterGen + 8 Paper2Poster).

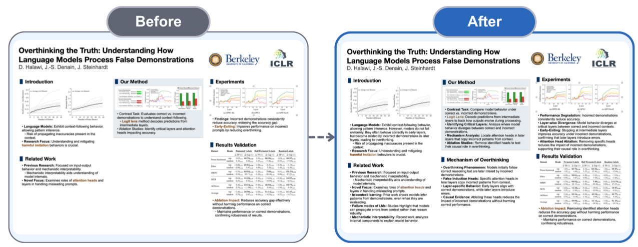
Stage	Taxonomy	Cases before	Cases after	Resolution
Stage 1	Overflow / out-of-bounds	8	0	1.00
	Overlap / collision	18	4	0.78
Stage 2	Whitespace / blank-region imbalance	14	6	0.57
	Section allocation / layout imbalance	11	1	0.91
Stage 3	Content insufficiency	16	7	0.56
	Content redundancy / noise / overload	12	5	0.58
Average repair resolution		–	–	0.73

Editable refinement benchmark. The editable optimization track evaluates whether staged refinement reduces active deficiencies without breaking the poster. For each executable taxonomy class, we manually count the number of active cases before refinement and the number that remain unresolved after refinement, then report the resulting per-class resolution rate. In the current workshop version, these counts are manually verified; future versions can scale this process with more automated evaluation tooling. Additional details are provided in Appendix A.5.

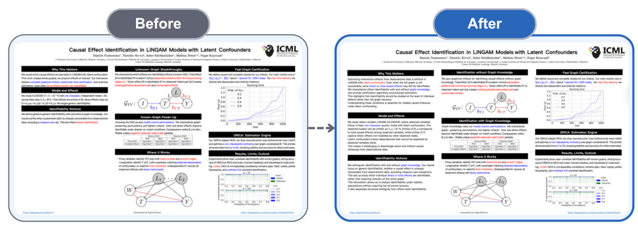
Holistic VLM-based evaluation. To complement class-wise diagnosis and repair metrics, we include a pilot holistic before/after evaluation using a frontier VLM judge under a fixed scoring rubric. We currently report 8 matched before/after cases: 4 PosterGen and 4 Paper2Poster. For each case, the judge compares the before and after posters with access to the source paper title and abstract, evaluating overall quality, layout/readability, and paper-grounded communication jointly. Details of the current pilot protocol and future evaluation extensions are provided in Appendix A.7 and Appendix A.8. With the current numbers, PROS reaches a macro diagnosis F1 of 0.82 on the mixed 16-case subset, achieves its strongest repair performance on Stage 1 structural failures, and shows a +29.1% average holistic gain

Table 4. Pilot VLM-as-a-judge evaluation on all 8 matched before/after cases in the current editable subset. Full per-pair scores are provided in Appendix A.7.

Source	#Cases	Overall Before	Overall After	Overall Gain	Gain (%)
PosterGen	4	3.3	4.1	+0.8	23.4%
Paper2Poster	4	2.8	3.9	+1.1	36.1%
Overall average	8	3.1	4.0	+0.9	29.1%



(a) PosterGen draft before and after three-stage refinement



(b) Paper2Poster draft before and after three-stage refinement

Figure 2. Representative before/after refinement examples from PosterGen and Paper2Poster. PROS improves layout-level issues such as whitespace and section balance, while also strengthening paper-grounded communication through clearer mechanism explanation and evidence emphasis.

across the currently scored 8 matched cases. These results are intended as preliminary evidence of feasibility rather than final benchmark performance.

4. Conclusion

PROS reframes poster improvement as proactive multimodal refinement rather than instruction execution alone. By coupling source-grounded diagnosis, staged repair, and grounded editing over editable poster states, it provides a practical starting point for human-interactive refinement of structured visual artifacts. On the current preliminary benchmark, PROS reaches a macro diagnosis F1 of 82.0%, an average repair resolution of 73.0%, and a pilot holistic gain of 29.1%. We view these as preliminary evidence of feasibility rather than final benchmark performance, but they suggest that proactive refinement is a promising direction for controllable editing when users cannot yet fully articulate what they need. This positions PROS as a human-interactive visual artifact editing loop that combines user-guided control with proactive multimodal refinement.

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A. Appendix

A.1. Dataset Construction

Our broader benchmark development is organized around *poster–paper pairs*, where each poster is paired with its original source paper. The current collection draws papers from four major venues—ICML, ICLR, CVPR, and NeurIPS—across 2024 and 2025, with coverage balanced across the two years.

The current data pool contains three categories: 36 **PosterGen-generated drafts**, 36 **Paper2Poster-generated drafts**, and 36 **human-made posters** collected from conference materials.

For the generated subsets, we run the corresponding PosterGen and Paper2Poster generation pipelines on selected source papers and use the resulting generated poster drafts in our benchmark, rather than treating prior released outputs as static artifacts. All three categories therefore share a consistent paired structure in which each poster is linked to its original paper.

A.2. Taxonomy Construction and Workshop Scope

We use these poster–paper pairs for two related purposes. First, generated drafts and human-made posters jointly support deficiency analysis and taxonomy construction. We conduct both human annotation and VLM-assisted annotation over these materials, then consolidate the most recurring and operationally meaningful issues into a broader 18-class deficiency taxonomy.

Second, the current workshop paper evaluates the 6-class executable core used in the main text. This subset is selected to span structural validity, layout-level readability, and source-grounded scientific communication while satisfying four operational criteria: recurrence in generated drafts, observability from poster–paper inputs, repairability over editable poster states, and evaluability through before/after deficiency resolution. The remaining classes are retained as part of the broader diagnosis inventory and can support future repair policies as their operational definitions and edit actions become more reliable.

A.3. Full 18-Class Deficiency Taxonomy

Table 5 summarizes the broader 18-class deficiency inventory underlying PROS. The six boldfaced classes are the executable subset used in the current workshop evaluation.

Table 5. Full 18-class deficiency taxonomy underlying PROS. The current workshop version centers on the six boldfaced classes, which are the most stable for executable refinement and quantitative evaluation.

Stage	Taxonomy class	Brief description
Stage 1	**Overflow / out-of-bounds**	Content extends beyond the poster canvas or designated container boundaries.
	Overlap / collision	Two or more content blocks intersect or visually collide.
	Clipping	Text or figures are partially cut off and become incomplete.
	Tiny-font failure	Font size is too small for normal poster viewing distance.
	Local clutter / congestion	Local regions are overly dense, crowded, or tightly packed.
	Pathological blank region	A large contiguous blank region reflects layout failure rather than functional whitespace.
Stage 2	**Excessive whitespace / blank-region imbalance**	The poster contains too much non-functional empty space or visibly imbalanced blank regions.
	Section allocation / layout imbalance	Visual area allocation across sections does not match structural importance or produces major layout imbalance.
	Evidence under-allocation / figure under-utilization	Key figures, tables, or evidence blocks occupy too little space or lack salience.
	Weak hierarchy / scan-path ambiguity	Visual hierarchy is too flat, making reading order and emphasis unclear.
	Claim emphasis failure	The strongest claim is present but is not made visually or informationally prominent.
	Coverage–emphasis mismatch	Covered content and emphasized content are misaligned in relative importance.
Stage 3	**Content insufficiency**	The poster omits or under-develops important source-paper content needed to understand the problem, method, evidence, or contribution.

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Table 5 continued from previous page

Stage	Taxonomy class	Brief description
	Content redundancy / noise / overload	The poster includes redundant, low-priority, or overly dense content that weakens readability and distracts from the central scientific message.
	Under-content	The poster carries clearly insufficient information relative to the paper's contribution.
	Over-textualization	The poster contains too much text and too little effective visual communication.
	Narrative fragmentation	The poster fails to form a coherent scientific reading path from problem to method to evidence and takeaway.
	Layout–content mismatch	The visual area assigned to content does not align with the semantic importance of that content in the paper.

A.4. Current Experimental Subset

The current quantitative experiments use an editable subset of 16 model-generated poster–paper pairs: 8 PosterGen-generated drafts and 8 Paper2Poster-generated drafts.

We use this subset because the present workshop study focuses on grounded diagnosis-and-repair evaluation over editable machine-generated poster drafts. Human-made posters are currently used primarily to support taxonomy construction and broader deficiency analysis rather than the main editable refinement results.

A.5. Evaluation Protocol

Diagnosis evaluation. For the six executable taxonomy classes, we manually annotate each poster in the current experimental subset and treat this annotation as the gold standard. Each class is represented as a binary label indicating whether that deficiency is present. System outputs are likewise converted into binary class-level predictions, where a detected class is counted as 1 and an undetected class as 0. We then compute per-class precision, recall, and F1, and report macro-F1 as the average F1 across the six classes. The current workshop version uses manual annotation and verification; later versions can scale this process with more automated tooling. For each poster, annotators record class-level human reference annotations, system-detected deficiencies, and post-repair unresolved deficiencies; Tables 2 and 3 report the resulting aggregated precision, recall, F1, and resolution rates.

Repair evaluation. For each executable class t, we manually count the number of active cases before repair, denoted N_t^{pre}, and the number of cases that remain unresolved after repair, denoted N_t^{post}. Resolution rate is then computed as

$$R_t = \frac{N_t^{pre} - N_t^{post}}{N_t^{pre}}.$$

The current workshop version uses manual counting and verification for these class-level repair statistics. In later versions, this process can likewise be scaled with more automated evaluation tooling.

A.6. Executable Diagnosis and Repair Details

Table 6 summarizes the diagnosis signals and repair actions used for the six executable taxonomy classes in the current workshop version. These details are intended to make the diagnosis-to-repair loop concrete without expanding the main text beyond the workshop scope.

Table 6. Executable diagnosis and repair details for the six-class workshop taxonomy.

Taxonomy class	Diagnosis signal	Repair action
Overflow / out-of-bounds	Detect text boxes, figures, or shapes whose bounding boxes extend beyond the poster canvas or their intended container regions.	Shrink, reposition, or resize the offending object or its container while preserving the local reading order and visual grouping.
Overlap / collision	Detect non-trivial intersections between text, figure, title, or section bounding boxes after rendering the poster and mapping visual regions back to editable objects.	Separate intersecting objects by moving, resizing, or locally reflowing neighboring elements while keeping section membership stable.

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Taxonomy class	Diagnosis signal	Repair action
Excessive whitespace / blank-region imbalance	Estimate large non-functional blank regions and column-level whitespace imbalance from rendered poster regions and section boundaries.	Expand nearby content, redistribute section area, or move adjacent blocks to reduce unused space without creating new overlap or overflow.
Section allocation / layout imbalance	Compare section area, column usage, and visual prominence against the poster's semantic structure and expected scientific reading flow.	Rebalance section heights, reposition sections, or adjust content allocation so that visually dominant regions better match communicative importance.
Content insufficiency	Compare poster section content against the source paper's problem, method, evidence, and contribution to identify missing or under-developed information.	Add or revise concise source-grounded content so that the poster more clearly communicates the necessary problem, method, evidence, or contribution.
Content redundancy / noise / overload	Detect redundant, low-priority, overly verbose, or weakly grounded content that increases reading burden without improving scientific communication.	Remove, compress, or reorganize redundant and noisy content while preserving the central scientific message and the poster's visual balance.

A.7. Holistic VLM-based Before/After Evaluation

To complement class-wise diagnosis and repair metrics, we include an auxiliary holistic before/after evaluation on 8 currently scored matched editable cases: 4 PosterGen cases and 4 Paper2Poster cases. For each case, the judge is given the source paper title and abstract together with the before and after poster images. We use a frontier VLM judge under a fixed scoring rubric for this pilot evaluation. The judge is prompted with the same fixed rubric for all cases and assigns a continuous holistic score on a 1–5 scale, where higher scores indicate stronger overall poster quality. The score jointly reflects layout/readability, visual organization, and paper-grounded communication. Percentage gain is computed as $(\text{After} - \text{Before})/\text{Before} \times 100\%$.

The judge evaluates three dimensions:

- **Overall quality**: which version is the better scientific poster overall;
- **Layout / readability**: which version has better spacing, balance, visual organization, and ease of scanning;
- **Paper-grounded communication**: which version more clearly and faithfully communicates the source paper's problem, method, and supporting evidence.

For each dimension, the judge produces before and after scores, which are then averaged over the sampled cases to obtain the gains reported in Table 4. Table 7 reports the per-pair overall scores underlying the aggregate results in Table 4.

Table 7. Per-pair VLM-as-a-judge before/after scores for the 8 matched editable cases.

Generation model	Paper / poster topic	Overall Before	Overall After	Gain	Gain (%)
PosterGen	LiNGAM latent confounders	3.4	4.2	+0.8	23.5%
	CORMAB / SEQUOIA	3.5	4.1	+0.6	17.1%
	VideoScene	3.3	4.0	+0.7	21.2%
	General-Reasoner	3.2	4.0	+0.8	25.0%
Paper2Poster	Label augmentation	2.6	3.8	+1.2	46.2%
	Overthinking false demonstrations	2.8	4.1	+1.3	46.4%
	Stochastic interpolants	2.9	3.8	+0.9	31.0%
	Causal LLM routing	3.0	3.9	+0.9	30.0%
Overall	Average	3.1	4.0	+0.9	29.1%

This evaluation is intended as a compact artifact-level complement to the class-wise diagnosis and repair results rather than a replacement for them. Future versions can strengthen this pilot with reproducible API-based and open-weight VLM judges, as described in Appendix A.8.

A.8. Future Evaluation Extensions

Beyond the current preliminary benchmark, future versions of PROS can strengthen evaluation through complementary automated and human-centered protocols.

Reproducible VLM-based judging. The current holistic before/after evaluation can be extended with an API-based and reproducible VLM judging protocol. In this setting, each matched poster pair would be evaluated using fixed model names, fixed prompts, a fixed 1–5 scoring rubric, and saved model outputs. Suitable judges include strong API-based VLMs such as GPT-4o or Gemini, together with open-weight or open-source VLMs such as Qwen2.5-VL and LLaVA-OneVision. Each judge would receive the source paper title and abstract together with the before and after poster images, then score overall quality, layout/readability, and paper-grounded communication. Results could be reported as average before score, average after score, score gain, and agreement across VLM judges.

Human A/B evaluation. A small human evaluation can ask independent raters to compare randomized before/after poster pairs. Each rater would see the source paper title and abstract together with two anonymized poster versions labeled A and B, rather than before and after. Raters would judge overall quality, layout/readability, and paper-grounded communication. Results could be reported as refined-poster preference rate, average score gain, and optional inter-rater agreement.

Poster-author user study. A further user-centered evaluation can recruit students or early-stage researchers who are preparing scientific posters. Participants would first revise a generated poster using their usual workflow, such as manual editing in PowerPoint, LaTeX, or Overleaf-style tools, and then revise another comparable poster with access to PROS's staged diagnosis-and-refinement guidance. The study would compare the resulting before/after posters and collect participant feedback on whether the staged taxonomy helped them identify problems, decide what to fix, and produce clearer poster revisions.

Together, these extensions would test PROS through complementary automated judgments, independent reader preferences, and poster-author editing workflows.

A.9. Auxiliary Data and Future Expansion

The current benchmark already mixes machine-generated drafts and human-made posters to support taxonomy construction and problem analysis. Beyond the present workshop scope, a natural next step is to convert selected human-made conference posters from PDF or image formats into editable PPTX-style artifacts so that human-created drafts can be incorporated into supplementary repair-oriented evaluation alongside model-generated posters.

We also plan to expand the model-generated subset beyond the current PosterGen and Paper2Poster drafts by incorporating additional paper-to-poster generation systems and pipelines when editable or convertible outputs are available. This would allow future evaluations to test whether PROS generalizes across different poster-generation distributions rather than only two source pipelines.

We also view augmentation as a useful extension path, but not as arbitrary corruption. Rather than introducing random noise, future benchmark expansion can use taxonomy-guided perturbations that inject or amplify specific deficiency patterns. This would allow more controlled testing of whether PROS can detect and repair targeted classes of poster deficiencies under editable settings.

A.10. Supplementary Statements

Impact Statement

This work aims to improve the quality and usability of scientific communication artifacts by supporting structured poster refinement. Potential positive impacts include reduced poster preparation burden, better communication of scientific findings, and assistance for users who can recognize that a poster needs improvement but cannot yet determine the right edits to request. Framed as human–AI collaboration, the system is intended to support authors both when they already know what they want to change and when they need help identifying and operationalizing improvement needs. Potential risks include over-automation of scientific summarization, refinements that inadvertently distort emphasis, or over-reliance on system-suggested repairs in settings where authorial intent matters. Our framework therefore emphasizes source-grounded editing, validity preservation, and human-inspectable staged refinement rather than unrestricted rewriting.

LLM/Agent Usage Disclosure

Large language models and agent-based tools were used in a strictly limited support capacity for language polishing, brainstorming alternatives, implementation assistance, and the auxiliary VLM-as-a-judge evaluation described in this paper. They did not generate the scientific contributions of this paper and were not used to determine the problem formulation, technical framework, taxonomy, evaluation design, interpretation of findings, or the paper's claims. All substantive intellectual contributions were made by the authors, who manually reviewed, validated, and revised any model-assisted material before inclusion. The authors retain full responsibility for the final submission.